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Letter of Recommendation

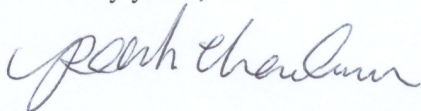
To whom it may concern,

It is with great pleasure that I write this recommendation letter for Mr. Iván López-Mata, born on June 20, 2002 in Cunduacan, Tabasco, Mexico. Mr. Lopez-Mata participated in the Excellence Cluster RESOLV internship program as a student in the Theoretical Chemistry Department at Ruhr University Bochum from May 14, 2024, to August 10, 2024. During his internship, Mr. Lopez-Mata actively contributed to our research project titled "Deciphering the Platinum Electrolyte Interface in Aqueous Solution." His specific contributions included:

- Calculating the optimal lattice parameter for platinum (Pt) using different functionals.
- Constructing Pt bulk slabs.
- Computing the density profiles of water molecules near the Pt interface.

Throughout the internship period, Mr. Lopez-Mata demonstrated exceptional motivation and passion for the project. His work was instrumental in setting up a Pt(111)/water interface. He optimized the lattice constant of Pt using various functionals and dispersion corrections and also computed the cohesive energies. Additionally, Mr. Lopez-Mata set up multiple slab geometries and performed ab initio molecular dynamics simulations for the Pt(111)/water interface, successfully computing the density profile of water near the interface. Mr. Lopez-Mata's strong background in programming and his ability to handle data efficiently in a high-performance computing cluster proved to be an asset to the project. His excellent communication and problem-solving skills allowed him to integrate seamlessly into the theory group, contributing valuable insights and results. In summary, Mr. Lopez-Mata's dedication, technical expertise, and teamwork made a significant impact on our project. I have no hesitation in strongly recommending him for future academic or professional opportunities.

Sincerely yours,



Dr. Chanbum Park
